

Gleysols

Gleysols are soils characterized by the presence of **gleyic properties**—that is, features caused by prolonged or periodic saturation with water, often because the groundwater table is near or at the soil surface for part of the year. These properties include greyish, bluish, or greenish colours in the matrix, mottles of oxidised iron/manganese, and reduced organic decomposition. They typically occur in **low-lying landscapes**, flat floodplains, depressions, wetlands, or places with poor drainage.

Drainage:

- **Wet season:** Very poor to imperfect drainage; soils become saturated, waterlogging is common, and oxygen deficiency sets in. Gleyic conditions (reduced iron, grey colours) develop.
- **Dry season:** Some desaturation occurs; surfaces may dry and develop cracks or fine fissures. However, water may still be held in the lower parts of the profile; drying is slower than in non-gleyed soils.
- **Landscape:** Found in floodplains, river basins, shallow swales, coastal flats, depressions, or any areas where the water table is periodically high or water accumulates.

Depth:

- Many Gleysols are **deep** soils (often >100 cm) though effective rooting depth may be constrained by persistent saturation, reduced sublayers or impeded aeration.
- In some cases, the gleyic properties extend close to the surface (within 50 cm) which influences many of the soil functions.

Workability:

- Soil structure can be easily degraded if tillage is done when the soil is too wet; saturated soils smear, puddle, and compact.
- When partly dry, workability improves, but the risk remains of damaging structure or leaving compaction.
- Because of the high water content and potential for anaerobic conditions, root growth and biological activity (worms, microbes) are often suppressed in wet periods.

Nutrient Richness & Cation Exchange Capacity (CEC):

- Gleysols often have **moderate to high CEC** (relative to many well-drained soils in similar climates), especially where fine texture, silt and clay are present. Organic matter tends to accumulate because decomposition is slower in saturated conditions.
- **Base saturation** can be variable: in lower parts or depressions they often receive inflows of base cations from adjacent uplands, improving base saturation; in others, leaching and acidity dominate.
- However, essential nutrients (especially phosphorus) may have low availability due to reductive conditions, iron/manganese interactions, and possible P fixation in wet or saturated layers. Nitrogen can be lost through denitrification under waterlogging.

pH:

- Tends to be slightly acidic to neutral in many Gleysols, though can become more acidic in strongly leached areas.
- In seasonal or fluctuating water regimes, pH shifts may occur due to redox cycles (release or immobilization of iron, manganese, mobilization of acidic compounds).

Management:

- **Drainage:** One of the most important interventions is to improve drainage (surface drains, ditches, raised beds, ridges) or to manage water table so that soil does not stay saturated for too long.
- **Land use:** Undrained Gleysols are often best suited for wetland or semi-aquatic crops (e.g. rice) or pastures, swamp forests. When drained, they can support a wider range of crops.
- **Organic matter management:** Maintain or increase surface organic matter (mulch, crop residues) to improve structure, reduce compaction, buffer cycles of wet/dry, and enhance microbial activity.
- **Fertilisation:** Balanced fertilisers especially N, P, K; special attention to phosphorus because availability may be low. Possibly use of liming in acidic Gleysols.
- **Avoid cultivating when wet:** Tillage when saturated causes structure loss, compaction, reduced permeability.
- **Crop selection:** Use water-tolerant or periodic waterlogging resistant crops where drainage is imperfect. Use raised beds or ridges in crop planting where possible.